Nexidia Real-Time Grid Artemis Setup

Version 12.5.1

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For assistance, contact your local supplier or nearest NICE Customer Service Center:

EMEA Region (Europe, Middle East, Africa)
Tel: +972-9-775-3800
Fax: +972-9-775-3000
email: support@nice.com

APAC Region (Asia/Pacific)
Tel: +852-8338-9818
Fax: +852-2802-1800
email: support.apac@nice.com

North America
Tel: 1-800-663-5601
Fax: +201-356-2197
email: na_sales@nice.com

International Headquarters-Israel
Tel: +972-9-775-3100
Fax: +972-9-775-3070
email: info@nice.com

The Americas Region (North, Central, South America)
Tel: 1-800-6423-611
Fax: +720-264-4012
email: support.americas@nice.com

Israel
Tel: 09-775-3333
Fax: 09-775-3000
email: support@nice.com

France
Tel: +33-(0)1-41-38-5000
Fax: +33-(0)1-41-38-5001

Hong-Kong
Tel: +852-2598-3838
Fax: +852-2802-1800

United Kingdom
Tel: +44-8707-22-4000
Fax: +44-8707-22-4500

Germany
Tel: +49-(0)-69-97177-0
Fax: +49-(0)-69-97177-200

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Install Apache ActiveMQ Artemis

1. Visit https://activemq.apache.org/components/artemis/download/
   Go to ActiveMQ Artemis 2.31.0 section, select apache-artemis-2.31.0-bin.zip file to download.
2. Once downloaded, extract it to some convenient folder location,
   e.g.: C:\apache-artemis-2.31.0 referred as ${ARTEMIS_HOME} in the admin guide.

> **TIP** Avoid having 'space' character in the folder name.

3. Open the command prompt and navigate to ${ARTEMIS_HOME}\bin
   Create a Broker Instance using the command:

```
${ARTEMIS_HOME}\bin\artemis create rtg-message-broker --user=[username] --password=[password] --allow-anonymous
```

Keep the user credentials (username / password) safe as the same will be used for console login.

4. Open ${ARTEMIS_HOME}\bin\rtg-message-broker\etc\broker.xml file.
5. Find this entry <acceptor name="artemis">tcp://0.0.0.0:61616?.......>
   Change the address from 0.0.0.0 to the address of the Artemis server. Change port number from 61616 to 26010.
   Some environments may require multiple acceptor entries to support JMS/TLS. Each additional acceptor should use a consecutive port, i.e. 26011, 26012, etc.

If using JMS/TLS for one or more acceptor entries, enter the encrypted password for the Java keystore and truststore used by the Artemis Message Broker. See the Password Generator for how to encrypt the keystore and truststore passwords if necessary.

In the following example, port 26010 is used for unencrypted JMS, and 26011 is used for JMS/TLS:

XML

```xml
<acceptor name="tcp">tcp:/10.2.50.
1:26010?tcpSendBufferSize=1048576;tcpReceiveBufferSize=1048576;amqpMinLargeMessageSize=102400;protocols=CORE,AMQP,STOMP,HORN
</acceptor>

<acceptor name="rtgssl">tcp://10.2.50.1:26011?sslEnabled=true;keyStorePath=../etc/example.keystore.
ks;keyStoreType=JCEKS;keyStorePassword=[encrypted keystore password];trustStorePath=../etc/example.
truststore.ts;trustStorePassword=[encrypted truststore
password]];needClientAuth=false;enabledProtocols=TLSv1,TLSv1.1,TLSv1.
2;tcpSendBufferSize=1048576;tcpReceiveBufferSize=1048576;amqpMinLargeMessageSize=102400;protocols=CORE,AMQP,STOMP,HORNETQ,MQ
```

1. Next, navigate to ${ARTEMIS_HOME}\bin\rtg-message-broker\bin
   Install ActiveMQ Artemis to Windows Service using the command:

```
${ARTEMIS_HOME}\bin\rtg-message-broker\bin\artemis-service.exe install
```

5. Open Windows Services Manager and start ActiveMQ Artemis service.
6. Verify your Broker Instance by opening the console in the browser: http://localhost:8161/console
7. Put the user credentials created during broker instance setup and login. You should be able to see the ActiveMQ Artemis console.

Configure Artemis Username and Password Credentials

Use the username and password entered when creating the message broker instance

1. Add the following libraries to the ${ARTEMIS_HOME}\bin\rtg-message-broker\bin\lib\ directory, these files can be found in the RTG install lib directory:

```
bcprov-jdk18on-1.76.jar
commons-codec-1.16.0.jar
nexidia-rtg-shared-12.5.0.0.jar
```

2. Open the ${ARTEMIS_HOME}\bin\rtg-message-broker\etc\broker.xml file and add the following properties under the <core> XML section

```
<mask-password>true</mask-password>
<password-codec>com.nexidia.rtg.artemis.
RtgCodec;keyStorePath=[keystoreFile];keyStoreType=JCEKS;keyStorePasscode=[keystorePasscode]</password-
codec>
```

where
[keystoreFile] is the path to the keystore file that contains the encryption key generated by the Password Generator
[keystorePasscode] is the generated passcode for the keystore file

3. Open the ${ARTEMIS_HOME}\bin\rtg-message-broker\etc\login.config, remove the GuestLoginModule to prevent anonymous logins, and make sure the PropertiesLoginModule exists

```
activemq {
  org.apache.activemq.artemis.spi.core.security.jaas.PropertiesLoginModule required
    debug=false
    reload=true
    org.apache.activemq.jaas.properties.user="artemis-users.properties"
    org.apache.activemq.jaas.properties.role="artemis-roles.properties";
};
```

4. Run the Password Generator to create the encrypted password and save the encrypted password for the next step
5. Open the command prompt and navigate to ${ARTEMIS_HOME}\bin\rtg-message-broker\bin directory and run the command

```
artemis mask --hash [encryptedPasswordFromPasswordGenerator]
```

6. Open the ${ARTEMIS_HOME}\bin\rtg-message-broker\etc\artemis-users.properties file and replace the user’s current encoded password with the new encoded password generated from the Artemis mask command

Configure Artemis Logging to write to the Encrypted Directory

1. Open the ${ARTEMIS_HOME}\bin\rtg-message-broker\etc\logging.properties file
2. Set the property handler.FILE.fileName to the encrypted directory path

> **handler.FILE.fileName=\\\\encrypt_dir\\Nexidia\\Real-Time Grid\\message-broker\\logs\\artemis.log**
>
> When using a network path, the "\" must be used to escape the backslash ("\") character

Password Generator

A command line tool used to generate an encryption key in the keystore file and encrypt passwords stored in the properties files.

1. Navigate to the RTG installation bin directory. The default location is `C:\Program Files\Nexidia\Real-Time Grid 12.5\bin`. Open a command prompt in the directory.
2. Run the command password-generator.cmd with the following parameters:

```
-keystoreFilename <keystore_filename>
-keystorePassphrase <keystore_password>
-keystoreType <keystore_type>
-generateEncryptionKey
-showEncryptedPassphrase
-passwords <password_list>
```

<*keystore_filename*> is the path of the keystore file where the encryption key exists. The file must exist. Required

<*keystore_password*> is the password of the keystore file. Required

<*keystore_type*> is the keystore type. The choices are JCEKS (Default) or PKCS12. Optional

-generateEncryptionKey generates the encryption key and stores it in the file defined by the <*keystore_filename*> value in the **-keystoreFilename** parameter. Optional

-showEncryptedPassphrase outputs the keystore password defined by the <*keystore_password*> value in the *-keystorePassphrase parameter. Optional

<*password_list*> comma delimited list of passwords or strings to encode. Optional

3. The output will show each password and its encrypted value.

The following example would encode a SQL connection string:

```
password-generator.cmd -keystoreFilename ./example.keystore -keystorePassphrase [unencrypted keystore password] -keystoreType JCEKS -generateEncryptionKey -showEncryptedPassphrase -passwords jdbc:sqlserver://example.sql.server.addre...xamplepassword

Passphrase encrypted: [encrypted keystore password] Passwords Encrypted: jdbc:sqlserver://example.sql.server.addre...xamplepassword: [encrypted SQL connection string]
```